

VI Semester B.E. Examination
(Common to ECE and ECE-II)
Automotive Electronics (24EECC305/24EEIC316)

Duration: 3 hours**Max. Marks: 100**

Note: i) Answer any TWO full questions from UNIT-I, any TWO full questions from UNIT-II and any ONE full question from UNIT-III.

ii) Use of steam table is permitted

iii) Assume suitable data if necessary with justification

UNIT-I

- 1 a. A fuel injector delivers fuel at a rate of 0.003 kg/sec. If the injector is open for 0.4 sec, calculate the total fuel delivered. Explain how feedback control loops are implemented in modern engine control systems. (08 marks)
- b. A sudden detonation is detected in a turbocharged engine due to pre-ignition at high RPM. Propose an electronic control system response to manage this condition. (06 marks)
- c. What are the legal requirements governing automotive air pollutants released into the atmosphere? Discuss the impact in terms of Indian and global standards. (06 marks)
- 2 a. In the event of a Mass Air Flow (MAF) sensor malfunction as detected by the Engine Control Unit's (ECU) diagnostic functions, how can the engine control system utilize alternative sensor data such as signals from the Manifold Absolute Pressure (MAP) sensor, Throttle Position Sensor (TPS), and Intake Air Temperature (IAT) sensor to estimate the air mass flow rate and maintain effective engine performance? (08 marks)
- b. Apply Model-Based Design (MBD) to create automotive ECU software, adhering to the V-model design process for automotive systems. (06 marks)
- c. Discuss the design and operation of the fuel injector actuator. (06 marks)
- 3 a. Provide a detailed overview of the various segments within the automotive industry and explain the automotive supply and value chain. (08 marks)
- b. A six-cylinder engine with a displacement of 3.2 liters operates at stoichiometric AFR. Assuming MAF = 0.045 kg/s and injector flow rate = 0.015 kg/s, calculate injector pulse duration. Modify the duration for cold start conditions. (06 marks)
- c. Identify three key sensors involved in closed-loop fuel management and explain how they influence the adjustment of injector timing and fuel delivery duration. (06 marks)

UNIT-II

- 4 a. With respect to LIN protocol, answer the following:
 - i) Frame structure and scheduling
 - ii) Slave-to-slave communication
 - iii) Message ID filtering
 - iv) Master task synchronization
 - v) Example use case in body electronics
 (08 marks)
- b. What do you mean by over steering and under steering, with the necessary break circuit configuration demonstrate how electronic stability program (ESP) counteracts the over steering and understanding condition? (06 marks)
- c. Name three sensors used for closed-loop fuel control. Explain their role in adjusting injector timing and pulse width. (06 marks)
- 5 a. Answer the following related to CAN Protocol: (08 marks)
 - i) Perform CRC check on the data 1001101 using polynomial 1011.
 - ii) Explain dominant and recessive states.
 - iii) How does CAN handle message arbitration?

iv) How are long messages handled in CAN?

v) Propose an acceptance filter for IDs: 0x230, 0x234, 0x238, 0x23C, 0x320.

b. Explain the concept of traction loss due to over-braking in icy conditions. How does ABS reduce the risk? Compute wheel slip (%) for a vehicle moving at 80 km/h, if the wheel rotational speed corresponds to 65 km/h. (06 marks)

c. With the necessary speed response plots discuss the role of for PI controllers for cruise control system. (06 marks)

6 a. The following schedule is used in a FlexRay cluster: Static slots for brake control, dynamic slots for infotainment.

i) Draw the communication cycle

ii) Draw the FlexRay cluster with at least 2 ECUs.

Communication Schedule

	Slot	Node	Frame	Channel	Event
Static Segment	1	Node A	A1	A	
			A1	B	
	2	Node B	B1	A	
		Node C	C1	B	
	3	Node D	D1	A	
			D2	B	
	4	Node E	E1	A	
		Node A	A2	B	
	5	Node C	C3	A	
		Node B	B2	B	
Dynamic Segment	6	Node A	A4	A	⚡
		Node B	B3	B	
	7	Node C	C4	A	
		Node D	D3	B	
	8	Node D	D3	A	
		Node E	E2	B	
	9	Node B	B4	A	
		Node A	A5	B	
	10	Node E	E3	A	
		Node C	C5	B	

(08 marks)

b. Explain the concepts of ride quality and vehicle handling dynamics in automotive systems. Discuss how an electronically controlled suspension system optimizes the trade-off between occupant comfort and dynamic stability under varying driving conditions. (06 marks)

c. Estimate LIN message components for the given specified parameters.

Specified Parameters
Hexadecimal
Decimal
Binary

Identifier: 0x39

Data Field:
1. 0x76
2. 0xDA
3. 0x73
4. 0xC5
5.
6.
7.
8.

Checksum:
Classic
Enhanced

Data Rate: 10.4 kBit/s

Message Length (min): Bit
P0:

Message Length (max): Bit
P1:

Transmission Time (min): ms
PID:

Transmission Time (max): ms
Checksum:

(06 marks)

7 a. An OEM is designing an ADAS system with Lane Departure Warning (LDW). Perform a Hazard Analysis and Risk Assessment (HARA) and assign the ASIL level based on ISO 26262. (10marks)

- b. Discuss SAE levels of autonomous driving. Identify the key features and name a car manufacturer that has deployed SAE Level 3 technology. (10marks)
- 8
 - a. Explain the objectives and components of On-Board Diagnostics (OBD-II). A vehicle exhibits irregular idling and misfire. List likely DTCs and possible root causes. (10marks)
 - b. During cold cranking, the AFR is poorly maintained. List possible reasons and explain how the KWP2000 protocol helps in diagnosing and addressing these faults. (10marks)